

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	Work done = $65\,000 \times 32 = 2.08 \times 10^6 \text{ J}$		1		1	1	
		(ii)	Energy at B = $2\,600 \times 9.81 \times 42$ (1) or 1.07×10^6 seen Substitution into % efficiency = $\frac{1.07 \times 10^6 \times 100\%}{2.08 \times 10^6}$ (1) (no ecf) % efficiency = 51.5% (1) [accept 51% – 55%, or 0.51 – 0.55]	1	1				
					1		3	3	
	(b)		Work against resistive forces = $2\,800 \times 36$ (1) or 101 kJ seen ΔE_p using 30 m drop = -765 kJ (1) Substitution into work-energy theorem: KE gain = 765 kJ – 101 kJ = 664 kJ (1) $664\,000 = \frac{1}{2} \times 2\,600 \times v^2$ (1) [allow use of 765 kJ] $v = 22.6 \text{ m s}^{-1}$ (1) [765 kJ $\rightarrow$ 24.3 m s ⁻¹] Additional marking guidance • Whole drop (42 m) with resistive forces $\rightarrow 27.3 \text{ m s}^{-1} \rightarrow$ 4 marks • 30 m drop without resistive forces $\rightarrow 24.3 \text{ m s}^{-1} \rightarrow$ 3 marks • Whole drop without resistive forces $\rightarrow 28.7 \text{ m s}^{-1} \rightarrow$ 2 marks • Mixing force and energy $\rightarrow$ 0 • Use of $v^2 = u^2 + 2ax \rightarrow$ 0 • Dissipate energy =	1	1 1 1 1		5	5	
			Question 7 total	2	7	0	9	9	0